Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Machado and Sons Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
11706 E Keyes RD	Denair	Stanislaus	95316

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1941

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X019-X038-X017-XXXX

B. OPERATORS

Machado, John

Operator name: Machado, John

Telephone no.: (209) 668-7781 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11706 E Keyes RD	Denair	CA	95316

Machado, Michael

Operator name: Machado, Michael

Telephone no.: (209) 678-2558 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11706 E Keyes RD	Denair	CA	95316

C. OWNERS

Machado, John

Legal owner name: Machado, John

Telephone no.: (209) 668-7781 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11706 E Keyes RD	Denair	CA	95316

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	240	200	95	95
Number under roof	665	105	0	0	0	0
Maximum number	665	105	240	200	95	95
Average number	665	105	240	200	95	95
Avg live weight (lbs)	1,400	1,425	950	700		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,760.73 *tons per reporting period*

Total nitrogen from manure: 295,696.10 *lbs per reporting period*

After ammonia losses (30% loss applied): 206,987.27 *lbs per reporting period*

Total phosphorus from manure: 48,414.93 *lbs per reporting period*

Total potassium from manure: 128,227.45 *lbs per reporting period*

Total salt from manure: 337,260.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,478,400 *gallons*

Total nitrogen generated: 69,445.10 *lbs*

Total phosphorus generated: 6,470.25 *lbs*

Total potassium generated: 122,519.59 *lbs*

Total salt generated: 690,392.00 *lbs*

	10,478,400	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,478,400	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW Freestalls	Ground water
IW Gratton	Ground water
TID Lateral 2.5	Ground water
TID Lateral 3	Ground water
TID Main	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/14/2023	Solid commercial fertilizer *4-0-0*	24.40 *ton*	As-is	0.1	4.000000	0.000000	0.000000	0.000000
07/13/2023	Solid commercial fertilizer *UN 32*	18.06 *ton*	As-is	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	13,510.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	13,510.40	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Almonds	56	56	1	process wastewater	X019-X037-X015-XXXX
Bryant	22	22	1	manure	X045-X039-X008-XXXX
Field 1	40	40	2	process wastewater	X019-X038-X017-XXXX
Field 2	45	45	2	process wastewater	X019-X038-X009-XXXX
Field 3	20	20	2	process wastewater	X019-X038-X003-XXXX
Geer	29	29	2	manure	X045-X059-X003-XXXX X045-X059-X018-XXXX
George	67	67	2	manure	X024-X038-X014-XXXX X024-X043-X003-XXXX X024-X043-X012-XXXX
Gratton	59	59	2	manure	X024-X042-X004-XXXX X024-X043-X001-XXXX
Hake	17	17	2	process wastewater	X019-X035-X018-XXXX
Hickman	34	34	2	manure	X080-X011-X038-XXXX
Perry	18	18	1	manure	X019-X028-X022-XXXX
Quincy	31	31	2	manure	X045-X044-X053-XXXX
Swanson	19	19	1	manure	X019-X034-X005-XXXX
Totals for areas that were used for application	457	457	22		
Totals for areas that were not used for application					
Land application area totals	457	457	22		

B. CROPS AND HARVESTS

Almonds

Field name: Almonds

Almonds

02/01/1998: Almond Field Run

Crop: Almond Field Run Acres planted: 56 Plant date: 02/01/1998

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/07/2023	380.00 *ton*	As-is		14.2	12,300.00	930.00	20,800.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	301.00	51.10	325.50	0.00
Total actual harvest content	6.79	166.93	12.62	282.29	1,315.80

Bryant

Field name: Bryant

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	426.00 *ton*	As-is		70.8	3,620.00	760.00	930.00		6.21

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.36	140.19	29.43	36.02	702.25

Field 1

Field name: Field 1

Field 1

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	704.00 *ton*	As-is		65.4	4,080.00	970.00	10,700.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.60	143.62	34.14	376.64	1,473.68

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,151.00 *ton*	As-is		66.8	4,420.00	970.00	6,270.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.78	254.37	55.82	360.84	1,356.57

Field 2

Field name: Field 2

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	947.00 *ton*	As-is		79.4	2,880.00	1,260.00	11,300.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	21.04	121.22	53.03	475.60	927.72

Field 2

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,246.00 *ton*	As-is		67.4	4,600.00	780.00	6,800.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.69	254.74	43.19	376.57	1,372.04

Field 3

Field name: Field 3

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	364.00 *ton*	As-is		67.2	3,710.00	1,080.00	11,800.00		7.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.20	135.04	39.31	429.52	928.87

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	468.00 *ton*	As-is		55.0	4,910.00	900.00	4,030.00		3.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.40	229.79	42.12	188.60	716.04

Geer

Field name: Geer

10/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	405.00 *ton*	As-is		61.3	4,350.00	840.00	6,840.00		5.33

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.97	121.50	23.46	191.05	576.14

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	816.00 *ton*	As-is		68.2	4,130.00	840.00	4,430.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.14	232.42	47.27	249.30	1,127.43

George

Field name: George

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 67 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	1,078.00 *ton*	As-is		62.8	4,540.00	1,120.00	11,200.00		9.01

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.09	146.09	36.04	360.41	1,078.55

George

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 67 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	1,689.00 *ton*	As-is		67.8	4,090.00	900.00	5,860.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.21	206.21	45.38	295.45	1,071.48

Gratton

Field name: Gratton

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 59 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	941.00 *ton*	As-is		65.4	3,770.00	970.00	10,700.00		8.07

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.95	120.26	30.94	341.31	890.67

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 59 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	1,458.00 *ton*	As-is		64.8	4,470.00	1,080.00	5,870.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.71	220.92	53.38	290.12	1,148.21

Hake

Field name: Hake

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	287.00 *ton*	As-is		66.4	4,200.00	1,080.00	12,800.00		7.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.88	141.81	36.47	432.19	813.43

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	541.00 *ton*	As-is		70.2	3,900.00	780.00	6,400.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.82	248.22	49.64	407.34	1,574.25

Hickman

Field name: Hickman

10/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 10/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	606.00 *ton*	As-is		63.4	3,590.00	890.00	7,100.00		5.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.82	127.97	31.73	253.09	658.86

Hickman

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	684.00 *ton*	As-is		57.6	4,960.00	960.00	3,680.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.12	199.57	38.63	148.07	750.63

Perry

Field name: Perry

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	432.00 *ton*	As-is		67.6	4,200.00	800.00	5,200.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.00	201.60	38.40	249.60	1,026.43

Quincy

Field name: Quincy

Quincy

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	586.00 *ton*	As-is		70.2	3,670.00	750.00	8,940.00		5.81

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	256.00	132.80	0.00
Total actual harvest content	18.90	138.75	28.35	337.99	654.57

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	922.00 *ton*	As-is		66.8	3,820.00	920.00	4,880.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.74	227.23	54.73	290.28	809.69

Swanson

Field name: Swanson

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	505.00 *ton*	As-is		67.6	4,240.00	820.00	5,160.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.58	225.39	43.59	274.29	1,136.73

NUTRIENT BUDGET

A. LAND APPLICATIONS

Almonds - 02/01/1998: Almond Field Run

Field name: Almonds

Crop: Almond Field Run

Plant date: 02/01/1998

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	154.95	11.19	290.67	1,723.95	1,345,200.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	0.00	7,695,000.00 *gal*
Application event totals		154.95	11.19	290.67	1,723.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.91	7,695,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	22.40	0.00	0.00	0.00	
Application event totals		22.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.12	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	22.40	0.00	0.00	0.00	
Application event totals		22.40	0.00	0.00	0.00	

Almonds - 02/01/1998: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.12	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.12	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	13.34	6,885,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	12.55	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	12.55	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.55	

Bryant - 10/30/2022: Oats, silage-soft dough

Field name: Bryant

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Bryant - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	162.73	70.35	438.66	8,712.47	170.00 *ton*
Application event totals		162.73	70.35	438.66	8,712.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	0.00	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	25.81	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.81	

Field 1 - 10/24/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.34	6.59	171.34	1,016.22	566,400.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		91.34	6.59	171.34	1,016.22	

Field 1 - 10/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.01	2.81	36.93	185.52	141,600.00 *gal*
IW Freestalls	Ground water	0.71	0.00	0.00	13.33	180,000.00 *gal*
Application event totals		22.71	2.81	36.93	198.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.03	9.83	129.24	649.32	495,600.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	11.72	4,320,000.00 *gal*
Application event totals		77.03	9.83	129.24	661.03	

Field 1 - 06/14/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	133.23	12.58	230.72	1,299.82	778,800.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	16.11	5,940,000.00 *gal*
Application event totals		133.23	12.58	230.72	1,315.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Field 1 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.65	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.95	9.72	178.28	1,004.40	601,800.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	12.45	4,590,000.00 *gal*
Application event totals		102.95	9.72	178.28	1,016.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.95	9.72	178.28	1,004.40	601,800.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	12.45	4,590,000.00 *gal*
Application event totals		102.95	9.72	178.28	1,016.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	13.18	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	13.18	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	12.45	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.45	

Field 1 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	11.72	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Field 2 - 10/24/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.19	5.86	152.30	903.31	566,400.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		81.19	5.86	152.30	903.31	

04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.25	9.99	131.29	659.62	566,400.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	10.41	4,320,000.00 *gal*
Application event totals		78.25	9.99	131.29	670.04	

Field 2 - 06/14/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.81	11.69	214.40	1,207.91	814,200.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	14.97	6,210,000.00 *gal*
Application event totals		123.81	11.69	214.40	1,222.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.32	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.90	9.15	167.79	945.32	637,200.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	11.72	4,860,000.00 *gal*
Application event totals		96.90	9.15	167.79	957.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.51	8.64	158.47	892.80	601,800.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	11.07	4,590,000.00 *gal*
Application event totals		91.51	8.64	158.47	903.87	

Field 2 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	10.41	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	10.41	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	9.76	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	9.11	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	9.11	

Field 3 - 10/24/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.34	6.59	171.34	1,016.22	283,200.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		91.34	6.59	171.34	1,016.22	

Field 3 - 10/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	88.03	11.24	147.71	742.08	283,200.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	11.72	2,160,000.00 *gal*
Application event totals		88.03	11.24	147.71	753.79	

Field 3 - 06/14/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.12	11.43	209.74	1,181.65	354,000.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	14.65	2,700,000.00 *gal*
Application event totals		121.12	11.43	209.74	1,196.30	

06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.65	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.65	

Field 3 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	11.72	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	11.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.90	9.15	167.79	945.32	283,200.00 *gal*
Application event totals		96.90	9.15	167.79	945.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.78	8.00	146.82	827.16	247,800.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	10.25	1,890,000.00 *gal*
Application event totals		84.78	8.00	146.82	837.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	10.25	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	8.79	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	8.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	8.79	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	8.79	

Geer - 10/14/2022: Oats, silage-soft dough

Field name: Geer

Crop: Oats, silage-soft dough

Plant date: 10/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	33.04	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	33.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	159.76	69.06	430.65	8,553.42	220.00 *ton*
Application event totals		159.76	69.06	430.65	8,553.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	0.00	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Geer - 06/07/2023: Corn, silage

Field name: Geer

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	213.22	69.96	44.98	11,827.13	260.00 *ton*
Application event totals		213.22	69.96	44.98	11,827.13	

Geer - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	37.76	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	31.47	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	31.47	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	29.89	3,847,500.00 *gal*
Application event totals		0.00	0.00	0.00	29.89	

Geer - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	28.32	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	25.17	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	25.17	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Ground water	0.00	0.00	0.00	23.60	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	23.60	

George - 10/31/2022: Oats, silage-soft dough

Field name: George

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	163.44	70.66	440.58	8,750.72	520.00 *ton*
Application event totals		163.44	70.66	440.58	8,750.72	

George - 10/31/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.43	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	0.00	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

George - 06/17/2023: Corn, silage

Field name: George

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	220.08	72.21	46.42	12,207.34	620.00 *ton*
Application event totals		220.08	72.21	46.42	12,207.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	19.24	11,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

George - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	17.49	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.49	

07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	17.49	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.49	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	17.49	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.49	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	15.74	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.74	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	16.61	10,260,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.61	

George - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	15.74	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.86	9,180,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.86	

Gratton - 10/30/2022: Oats, silage-soft dough

Field name: Gratton

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	160.62	69.43	432.97	8,599.55	450.00 *ton*
Application event totals		160.62	69.43	432.97	8,599.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	0.00	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	16.38	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.38	

Gratton - 06/17/2023: Corn, silage

Field name: Gratton

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	221.70	72.75	46.76	12,297.44	550.00 *ton*
Application event totals		221.70	72.75	46.76	12,297.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	18.62	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	17.13	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	16.38	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.38	

Gratton - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	16.38	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.89	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	15.64	8,505,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	15.27	8,302,500.00 *gal*
Application event totals		0.00	0.00	0.00	15.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.89	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.89	

Hake - 10/24/2022: Oats, silage-soft dough

Field name: Hake

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Hake - 10/24/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.46	7.76	201.58	1,195.55	283,200.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		107.46	7.76	201.58	1,195.55	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.68	9.92	130.33	654.77	212,400.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	10.34	1,620,000.00 *gal*
Application event totals		77.68	9.92	130.33	665.11	

Hake - 06/16/2023: Corn, silage

Field name: Hake

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	128.24	12.10	222.08	1,251.16	318,600.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	15.51	2,430,000.00 *gal*
Application event totals		128.24	12.10	222.08	1,266.67	

06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Hake - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.65	2,295,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	13.78	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.75	9.41	172.73	973.13	247,800.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	12.06	1,890,000.00 *gal*
Application event totals		99.75	9.41	172.73	985.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.75	9.41	172.73	973.13	247,800.00 *gal*
TID Main	Ground water	0.00	0.00	0.00	12.06	1,890,000.00 *gal*
Application event totals		99.75	9.41	172.73	985.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	13.78	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	13.78	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	13.78	

Hake - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	12.06	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.06	

Hickman - 10/14/2022: Oats, silage-soft dough

Field name: Hickman

Crop: Oats, silage-soft dough

Plant date: 10/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	161.04	69.62	434.10	8,622.03	260.00 *ton*
Application event totals		161.04	69.62	434.10	8,622.03	

10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	0.00	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.21	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.21	

Hickman - 06/15/2023: Corn, silage

Field name: Hickman

Crop: Corn, silage

Plant date: 06/15/2023

Hickman - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	216.84	71.15	45.74	12,027.82	310.00 *ton*
Application event totals		216.84	71.15	45.74	12,027.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	15.51	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.21	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	14.21	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	14.21	

Hickman - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	15.51	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	13.57	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	13.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	12.92	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	12.92	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Main	Ground water	0.00	0.00	0.00	12.92	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	12.92	

Perry - 06/15/2023: Corn, silage

Field name: Perry

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Perry - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	211.40	69.37	44.59	11,726.04	160.00 *ton*
Application event totals		211.40	69.37	44.59	11,726.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	27.60	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	23.66	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	21.69	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	21.69	

Perry - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	19.72	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	19.72	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	15.77	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	15.77	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	15.77	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.77	

Quincy - 10/30/2022: Oats, silage-soft dough

Field name: Quincy

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Quincy - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	163.04	70.48	439.49	8,729.00	240.00 *ton*
Application event totals		163.04	70.48	439.49	8,729.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	27.47	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	27.47	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.47	

Quincy - 06/07/2023: Corn, silage

Field name: Quincy

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	214.81	70.48	45.31	11,915.17	280.00 *ton*
Application event totals		214.81	70.48	45.31	11,915.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	29.76	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.76	

Quincy - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	25.18	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.18	

07/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	25.18	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.18	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	22.89	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.89	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	22.89	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.89	

Quincy - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	20.61	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	18.32	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	18.32	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.32	

Swanson - 06/15/2023: Corn, silage

Field name: Swanson

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	212.79	69.82	44.89	11,803.19	170.00 *ton*
Application event totals		212.79	69.82	44.89	11,803.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	29.88	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.88	

Swanson - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	29.88	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.88	

07/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	26.15	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.15	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	29.88	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.88	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	29.88	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.88	

Swanson - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	26.15	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	22.41	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.41	

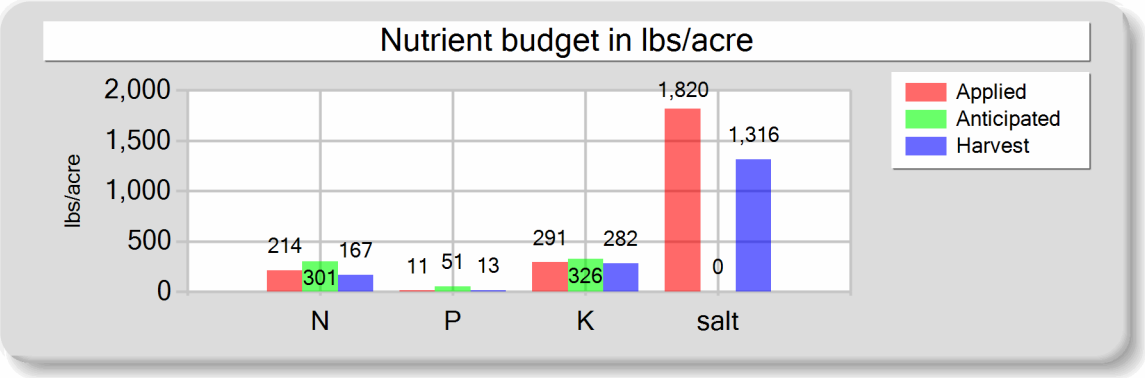
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Ground water	0.00	0.00	0.00	22.41	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.41	

B. NUTRIENT BUDGET

Almonds - 02/01/1998: Almond Field Run

Field name: Almonds Crop: Almond Field Run Plant date: 02/01/1998



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	44.80	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.95	11.19	290.67	1,723.95
Fresh water	0.00	0.00	0.00	95.72
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	213.75	11.19	290.67	1,819.66
Anticipated crop nutrient removal	301.00	51.10	325.50	0.00
Actual crop nutrient removal	166.93	12.62	282.29	1,315.80
Nutrient balance	46.83	-1.44	8.38	503.86
Applied to removed ratio	1.28	0.89	1.03	1.38

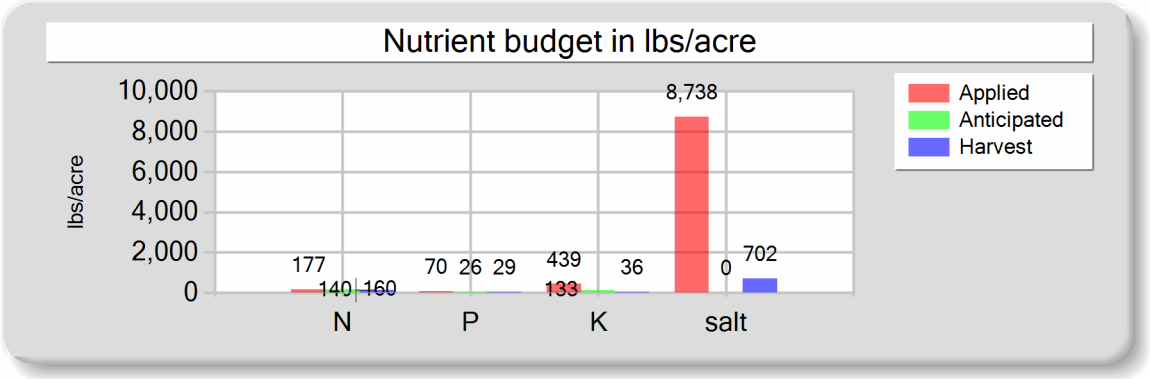
Fresh water applied	
57,105,000.00	*gallons*
2,102.98	*acre-inches*
37.55	*inches/acre*

Process wastewater applied	
1,345,200.00	*gallons*
49.54	*acre-inches*
0.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bryant - 10/30/2022: Oats, silage-soft dough

Field name: Bryant | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	162.73	70.35	438.66	8,712.47
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	25.81
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	176.73	70.35	438.66	8,738.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	140.19	29.43	36.02	702.25
Nutrient balance	36.53	40.91	402.64	8,036.03
Applied to removed ratio	1.26	2.39	12.18	12.44

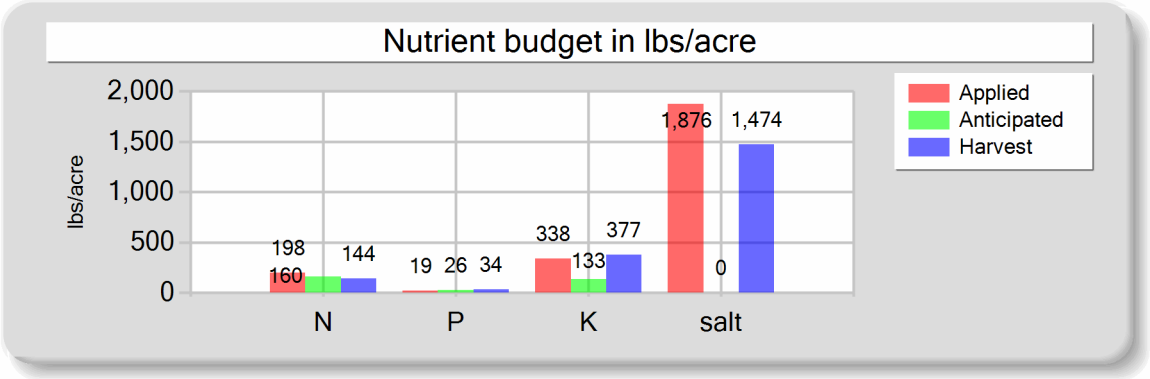
Fresh water applied	
6,480,000.00	*gallons*
238.64	*acre-inches*
10.85	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 10/24/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.38	19.24	337.51	1,851.06
Fresh water	0.71	0.00	0.00	25.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.08	19.24	337.51	1,876.11
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.62	34.14	376.64	1,473.68
Nutrient balance	54.47	-14.91	-39.13	402.42
Applied to removed ratio	1.38	0.56	0.90	1.27

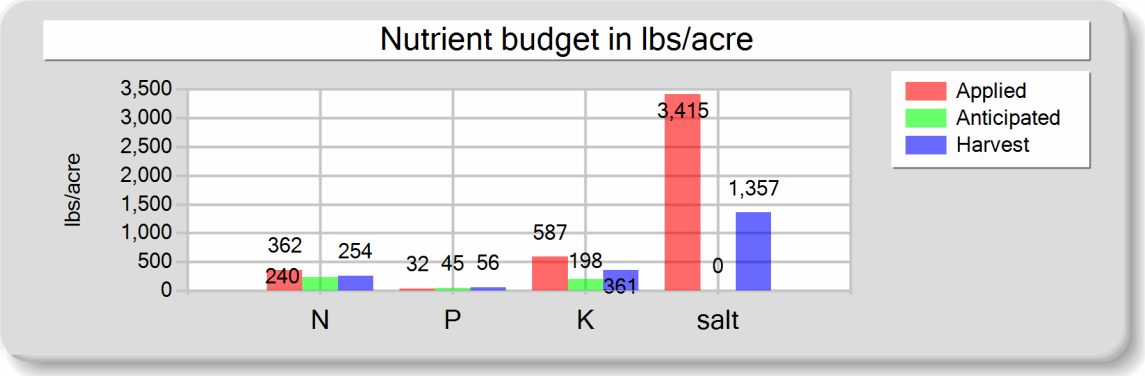
Fresh water applied	
8,820,000.00	*gallons*
324.81	*acre-inches*
8.12	*inches/acre*

Process wastewater applied	
1,203,600.00	*gallons*
44.32	*acre-inches*
1.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/14/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	339.13	32.01	587.28	3,308.63
Fresh water	0.00	0.00	0.00	106.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	362.13	32.01	587.28	3,414.81
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	254.37	55.82	360.84	1,356.57
Nutrient balance	107.76	-23.81	226.44	2,058.24
Applied to removed ratio	1.42	0.57	1.63	2.52

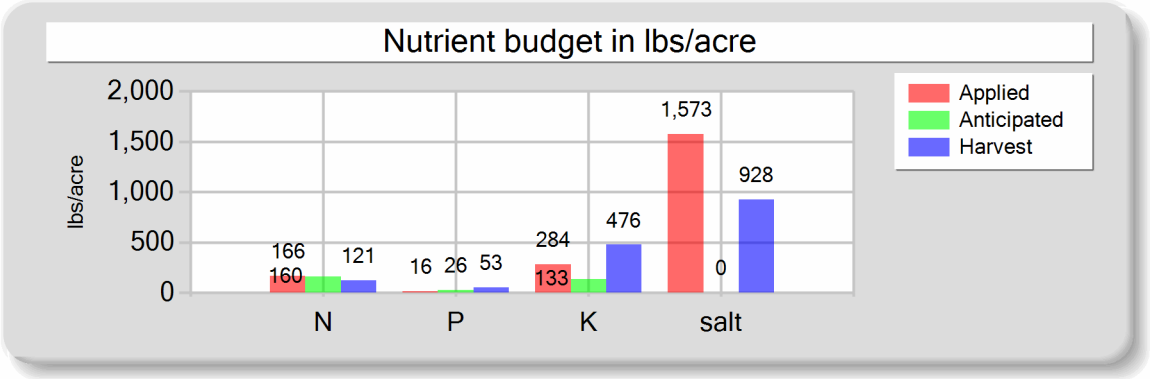
Fresh water applied	
39,150,000.00	*gallons*
1,441.76	*acre-inches*
36.04	*inches/acre*

Process wastewater applied	
1,982,400.00	*gallons*
73.01	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/24/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	159.44	15.85	283.60	1,562.93
Fresh water	0.00	0.00	0.00	10.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	166.44	15.85	283.60	1,573.35
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.22	53.03	475.60	927.72
Nutrient balance	45.23	-37.18	-192.01	645.62
Applied to removed ratio	1.37	0.30	0.60	1.70

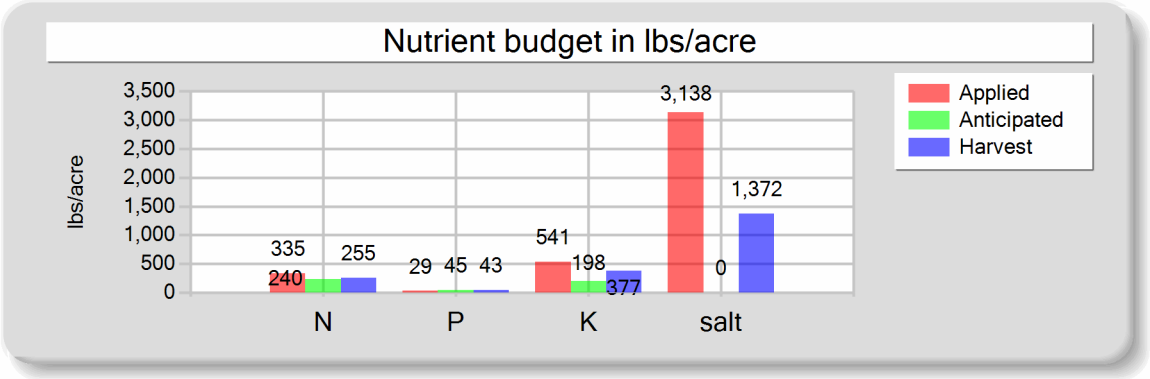
Fresh water applied	
8,640,000.00	*gallons*
318.18	*acre-inches*
7.07	*inches/acre*

Process wastewater applied	
1,132,800.00	*gallons*
41.72	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/14/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	312.22	29.47	540.67	3,046.04
Fresh water	0.00	0.00	0.00	91.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.22	29.47	540.67	3,137.81
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	254.74	43.19	376.57	1,372.04
Nutrient balance	80.48	-13.72	164.10	1,765.77
Applied to removed ratio	1.32	0.68	1.44	2.29

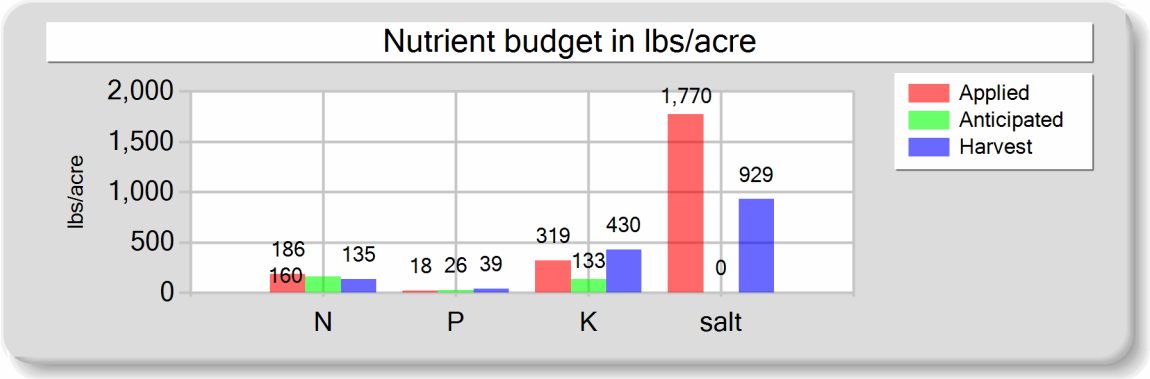
Fresh water applied	
38,070,000.00	*gallons*
1,401.99	*acre-inches*
31.16	*inches/acre*

Process wastewater applied	
2,053,200.00	*gallons*
75.61	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/24/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	179.37	17.83	319.05	1,758.30
Fresh water	0.00	0.00	0.00	11.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	186.37	17.83	319.05	1,770.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	135.04	39.31	429.52	928.87
Nutrient balance	51.33	-21.48	-110.47	841.14
Applied to removed ratio	1.38	0.45	0.74	1.91

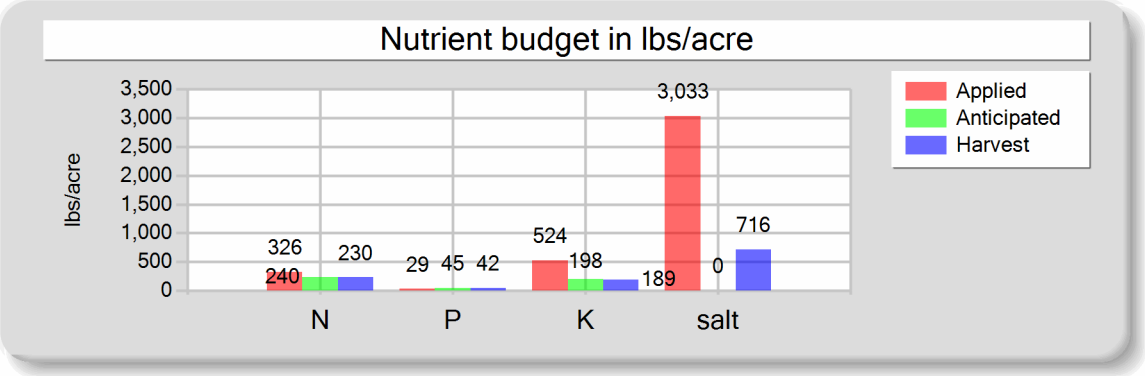
Fresh water applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
7.95	*inches/acre*

Process wastewater applied	
566,400.00	*gallons*
20.86	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/14/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	302.80	28.58	524.36	2,954.13
Fresh water	0.00	0.00	0.00	79.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.80	28.58	524.36	3,033.22
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	229.79	42.12	188.60	716.04
Nutrient balance	96.01	-13.54	335.75	2,317.18
Applied to removed ratio	1.42	0.68	2.78	4.24

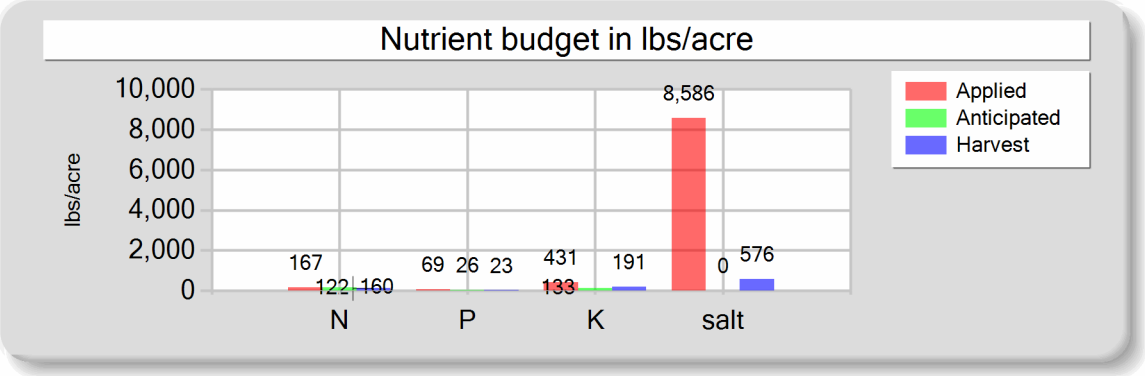
Fresh water applied	
14,580,000.00	*gallons*
536.93	*acre-inches*
26.85	*inches/acre*

Process wastewater applied	
885,000.00	*gallons*
32.59	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

Geer - 10/14/2022: Oats, silage-soft dough

Field name: Geer | Crop: Oats, silage-soft dough | Plant date: 10/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	159.76	69.06	430.65	8,553.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	33.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	166.76	69.06	430.65	8,586.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.50	23.46	191.05	576.14
Nutrient balance	45.26	45.60	239.60	8,010.32
Applied to removed ratio	1.37	2.94	2.25	14.90

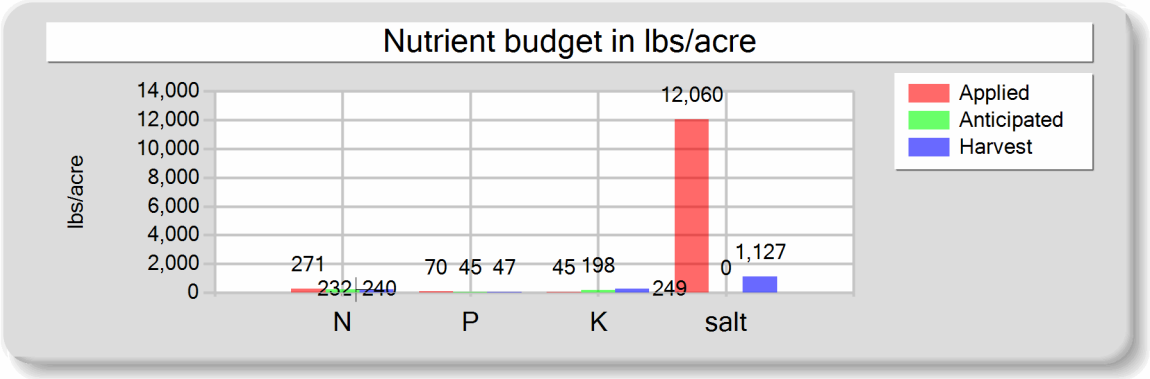
Fresh water applied	
8,707,500.00	*gallons*
320.67	*acre-inches*
11.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Geer - 06/07/2023: Corn, silage

Field name: Geer | Crop: Corn, silage | Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	213.22	69.96	44.98	11,827.13
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	232.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.42	69.96	44.98	12,059.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	232.42	47.27	249.30	1,127.43
Nutrient balance	39.00	22.69	-204.33	10,932.55
Applied to removed ratio	1.17	1.48	0.18	10.70

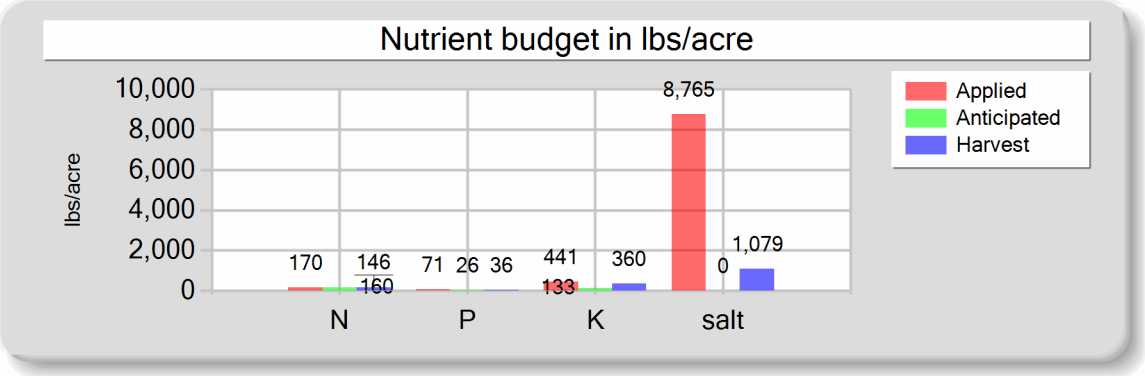
Fresh water applied	
29,970,000.00	*gallons*
1,103.69	*acre-inches*
38.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

George - 10/31/2022: Oats, silage-soft dough

Field name: George | Crop: Oats, silage-soft dough | Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.44	70.66	440.58	8,750.72
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	14.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.44	70.66	440.58	8,765.14
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.09	36.04	360.41	1,078.55
Nutrient balance	24.35	34.61	80.18	7,686.59
Applied to removed ratio	1.17	1.96	1.22	8.13

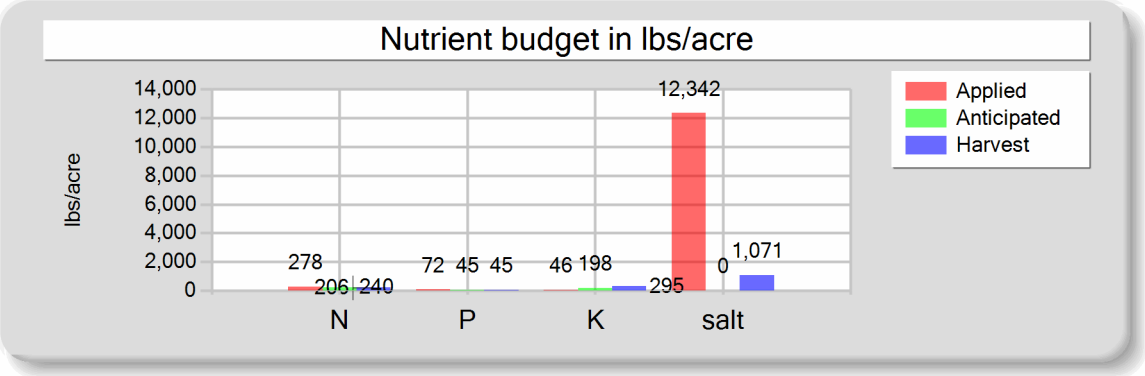
Fresh water applied
18,225,000.00 *gallons*
671.16 *acre-inches*
10.02 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

George - 06/17/2023: Corn, silage

Field name: George | Crop: Corn, silage | Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	220.08	72.21	46.42	12,207.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	134.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.28	72.21	46.42	12,341.99
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	206.21	45.38	295.45	1,071.48
Nutrient balance	72.07	26.84	-249.03	11,270.51
Applied to removed ratio	1.35	1.59	0.16	11.52

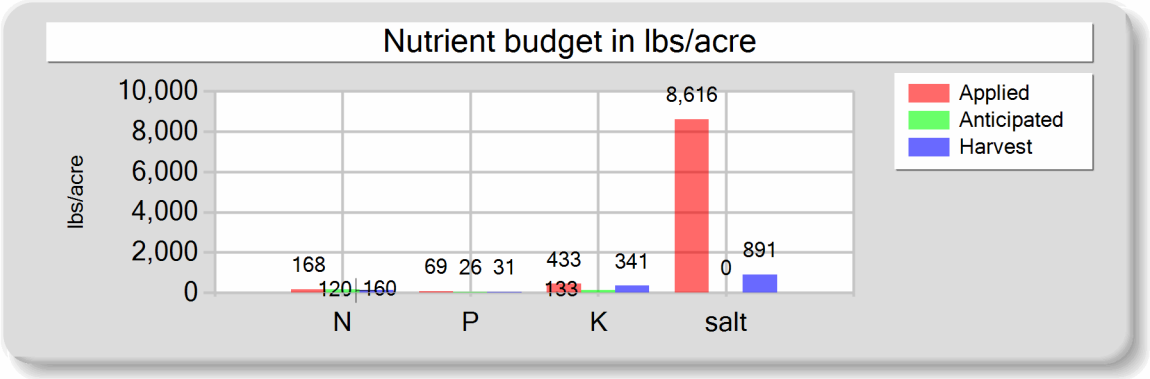
Fresh water applied	
83,160,000.00	*gallons*
3,062.50	*acre-inches*
45.71	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Gratton - 10/30/2022: Oats, silage-soft dough

Field name: Gratton | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	160.62	69.43	432.97	8,599.55
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	16.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	167.62	69.43	432.97	8,615.93
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	120.26	30.94	341.31	890.67
Nutrient balance	47.36	38.49	91.66	7,725.26
Applied to removed ratio	1.39	2.24	1.27	9.67

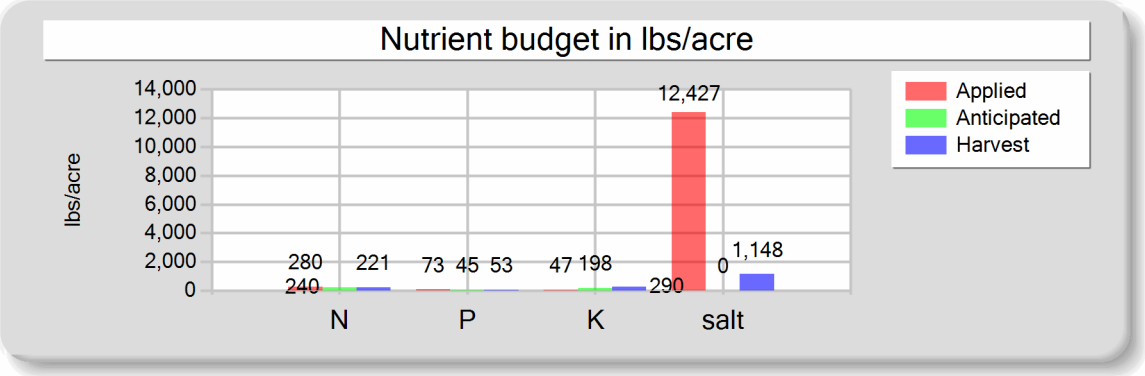
Fresh water applied	
18,225,000.00	*gallons*
671.16	*acre-inches*
11.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Gratton - 06/17/2023: Corn, silage

Field name: Gratton | Crop: Corn, silage | Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	221.70	72.75	46.76	12,297.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	129.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.90	72.75	46.76	12,426.64
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	220.92	53.38	290.12	1,148.21
Nutrient balance	58.98	19.37	-243.35	11,278.43
Applied to removed ratio	1.27	1.36	0.16	10.82

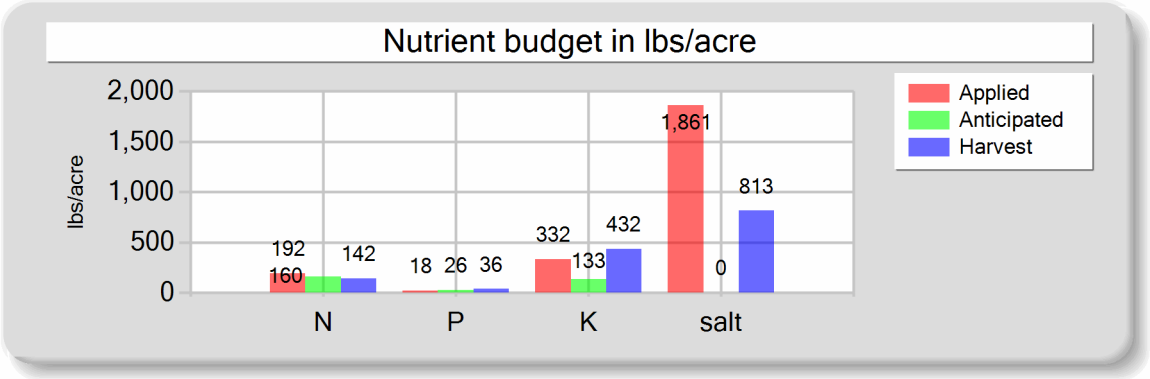
Fresh water applied	
70,267,500.00	*gallons*
2,587.71	*acre-inches*
43.86	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hake - 10/24/2022: Oats, silage-soft dough

Field name: Hake | Crop: Oats, silage-soft dough | Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	185.14	17.67	331.91	1,850.33
Fresh water	0.00	0.00	0.00	10.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.14	17.67	331.91	1,860.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.81	36.47	432.19	813.43
Nutrient balance	50.33	-18.79	-100.28	1,047.23
Applied to removed ratio	1.35	0.48	0.77	2.29

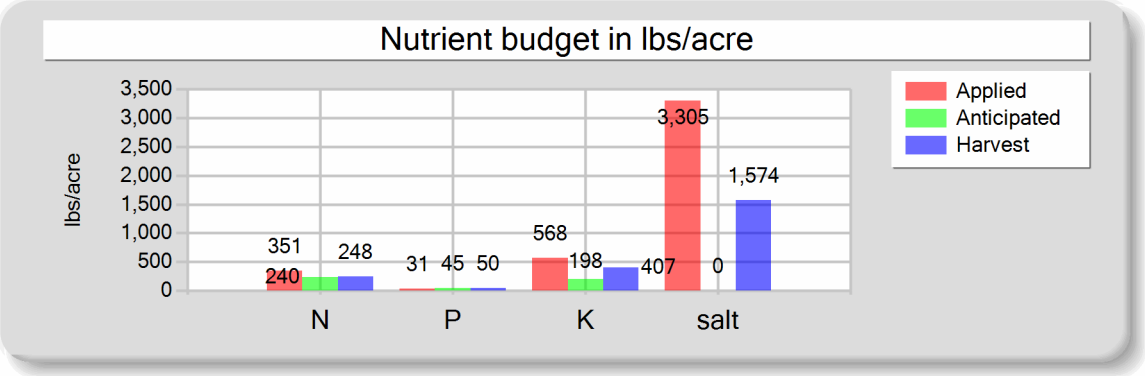
Fresh water applied	
3,780,000.00	*gallons*
139.20	*acre-inches*
8.19	*inches/acre*

Process wastewater applied	
495,600.00	*gallons*
18.25	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hake - 06/16/2023: Corn, silage

Field name: Hake | Crop: Corn, silage | Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	327.73	30.93	567.54	3,197.41
Fresh water	0.00	0.00	0.00	107.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.73	30.93	567.54	3,305.10
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.22	49.64	407.34	1,574.25
Nutrient balance	102.51	-18.71	160.20	1,730.85
Applied to removed ratio	1.41	0.62	1.39	2.10

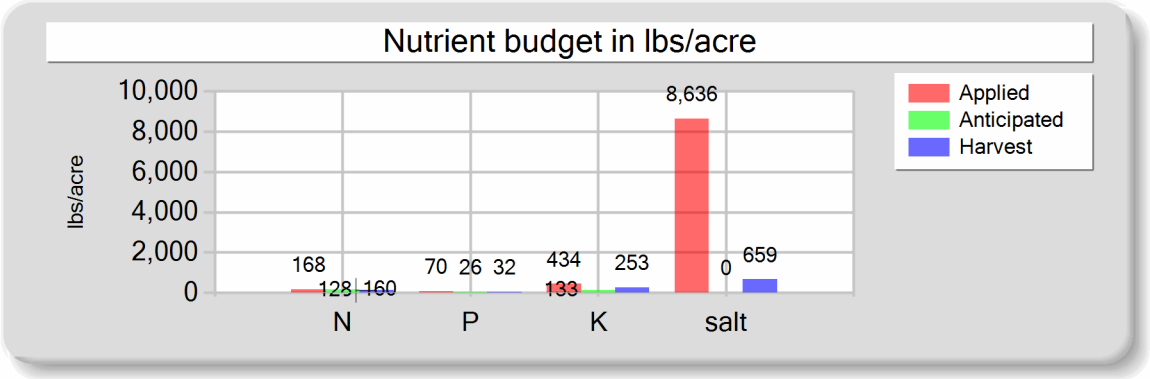
Fresh water applied	
16,875,000.00	*gallons*
621.45	*acre-inches*
36.56	*inches/acre*

Process wastewater applied	
814,200.00	*gallons*
29.98	*acre-inches*
1.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hickman - 10/14/2022: Oats, silage-soft dough

Field name: Hickman | Crop: Oats, silage-soft dough | Plant date: 10/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	161.04	69.62	434.10	8,622.03
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	14.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	168.04	69.62	434.10	8,636.24
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	127.97	31.73	253.09	658.86
Nutrient balance	40.06	37.89	181.01	7,977.38
Applied to removed ratio	1.31	2.19	1.72	13.11

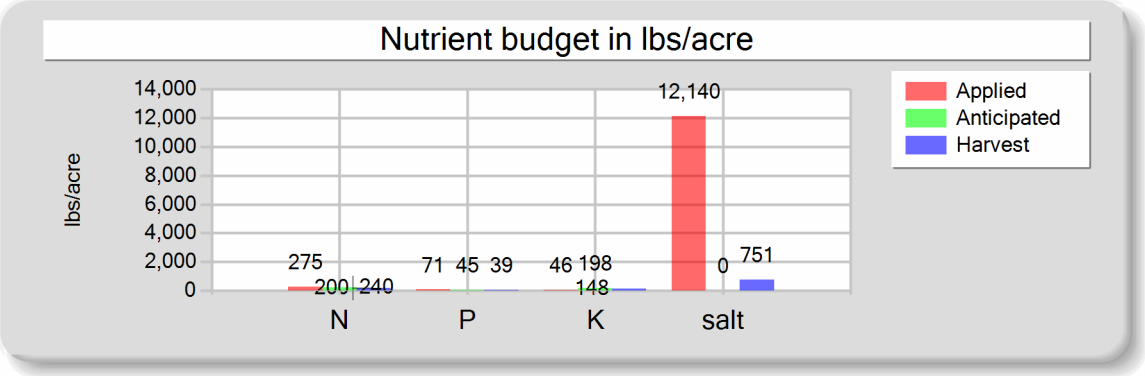
Fresh water applied	
9,315,000.00	*gallons*
343.04	*acre-inches*
10.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hickman - 06/15/2023: Corn, silage

Field name: Hickman | Crop: Corn, silage | Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	216.84	71.15	45.74	12,027.82
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	111.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.04	71.15	45.74	12,139.60
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	199.57	38.63	148.07	750.63
Nutrient balance	75.47	32.52	-102.33	11,388.97
Applied to removed ratio	1.38	1.84	0.31	16.17

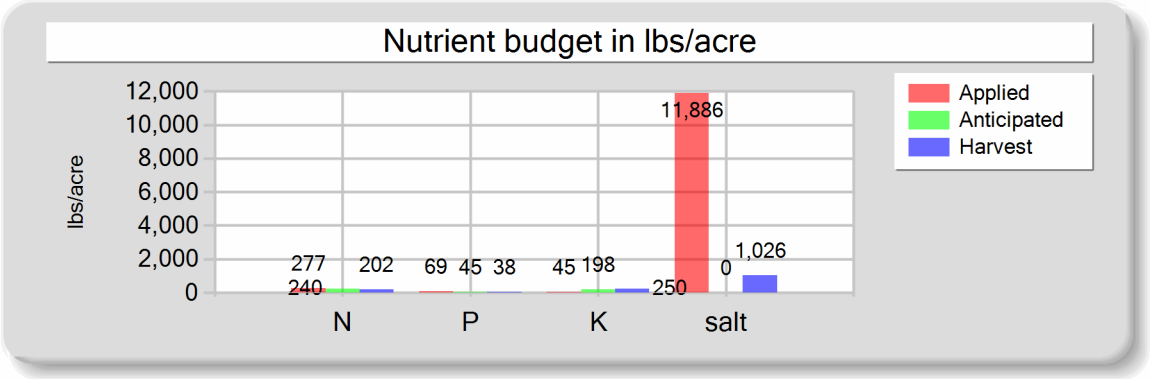
Fresh water applied	
35,032,500.00	*gallons*
1,290.13	*acre-inches*
37.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Perry - 06/15/2023: Corn, silage

Field name: Perry | Crop: Corn, silage | Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	211.40	69.37	44.59	11,726.04
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	159.69
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	276.60	69.37	44.59	11,885.74
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.60	38.40	249.60	1,026.43
Nutrient balance	75.00	30.97	-205.01	10,859.30
Applied to removed ratio	1.37	1.81	0.18	11.58

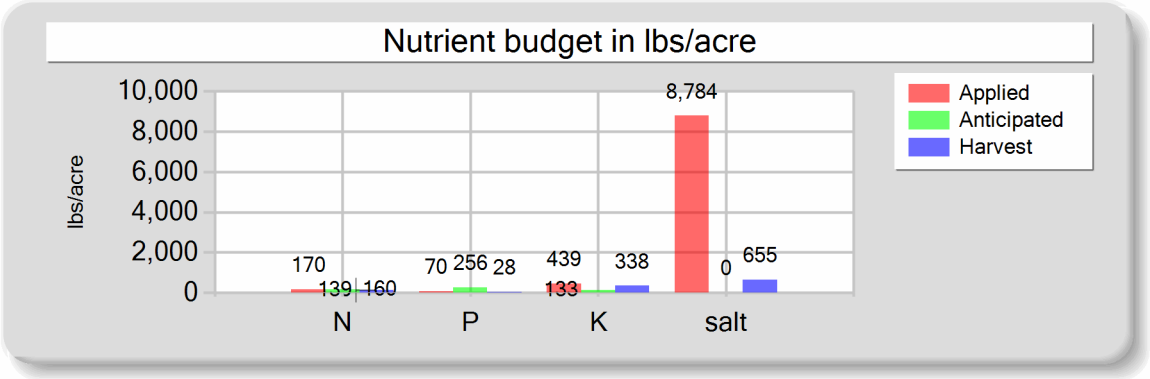
Fresh water applied	
16,402,500.00	*gallons*
604.05	*acre-inches*
33.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Quincy - 10/30/2022: Oats, silage-soft dough

Field name: Quincy | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.04	70.48	439.49	8,729.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	54.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.04	70.48	439.49	8,783.95
Anticipated crop nutrient removal	160.00	256.00	132.80	0.00
Actual crop nutrient removal	138.75	28.35	337.99	654.57
Nutrient balance	31.29	42.13	101.50	8,129.38
Applied to removed ratio	1.23	2.49	1.30	13.42

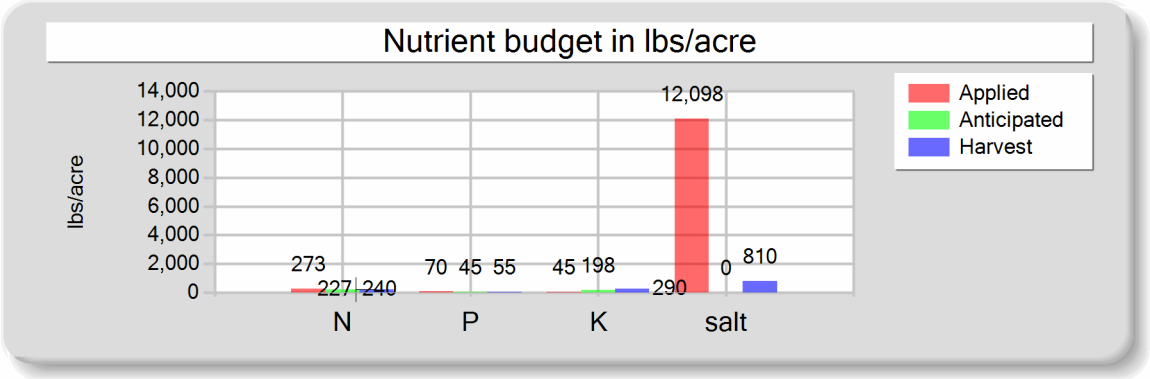
Fresh water applied	
9,720,000.00	*gallons*
357.95	*acre-inches*
11.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Quincy - 06/07/2023: Corn, silage

Field name: Quincy | Crop: Corn, silage | Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	214.81	70.48	45.31	11,915.17
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	183.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	273.01	70.48	45.31	12,098.33
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	227.23	54.73	290.28	809.69
Nutrient balance	45.78	15.76	-244.97	11,288.64
Applied to removed ratio	1.20	1.29	0.16	14.94

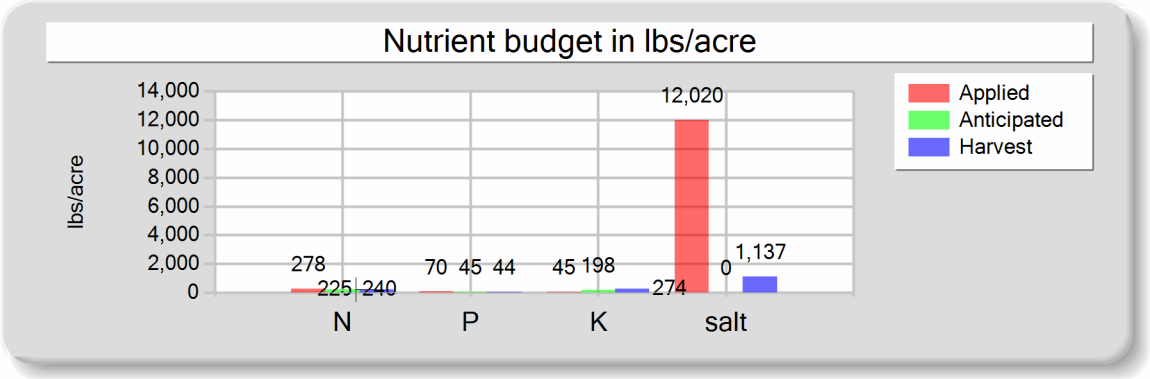
Fresh water applied	
32,400,000.00	*gallons*
1,193.18	*acre-inches*
38.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Swanson - 06/15/2023: Corn, silage

Field name: Swanson | Crop: Corn, silage | Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	212.79	69.82	44.89	11,803.19
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	216.66
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	277.99	69.82	44.89	12,019.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.39	43.59	274.29	1,136.73
Nutrient balance	52.60	26.23	-229.41	10,883.12
Applied to removed ratio	1.23	1.60	0.16	10.57

Fresh water applied	
23,490,000.00	*gallons*
865.06	*acre-inches*
45.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	6,960.00	43,400.00							86.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	4,200.00	2,700.00							71.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,100.00	4,520.00	23,900.00							81.40
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	773.00	412.00	0.00	0.00	55.80	1,450.00								9,610.00	8,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 01/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	745.00	448.00	0.00	0.00	95.10	1,250.00								8,440.00	6,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,880.00	638.00	0.00	0.00	264.00	1,500.00	231.00	196.00	221.00	5,860.00	0.00	279.00	195.00	9,280.00	4,980
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	50.00	1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	820.00	571.00	0.00	0.00	77.40	1,420.00								9,360.00	8,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW Freestalls

Fresh Water

Sample description: Fresh Water

Sample date: 12/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	18.80	0.00	18.80								655.00	355
DL	0.50	0.50	0.50								1.00	10

IW Gratton

Fresh Water

Sample description: Fresh Water

Sample date: 12/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.70	0.00	13.80								815.00	470
DL	0.50	0.50	0.50								1.00	10

TID Lateral 2.5

Fresh Water

Sample description: Fresh Water

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.50	21
DL	0.50	0.50	0.50								1.00	10

TID Lateral 3

Fresh Water

Sample description: Fresh Water

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								35.90	27
DL	0.50	0.50	0.50								1.00	10

TID Main

Fresh Water

Sample description: Fresh Water

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.30	13
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Almonds - 02/01/1998: Almond Field Run

Almonds - 02/01/1998: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 14.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	930.00	20,800.00		11.30
DL	100.00	100.00	100.00		0.01

Bryant - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,620.00	760.00	930.00		6.21
DL	100.00	100.00	100.00		0.01

Field 1 - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,080.00	970.00	10,700.00		12.10
DL	100.00	100.00	100.00		0.01

Field 1 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,420.00	970.00	6,270.00		7.10
DL	100.00	100.00	100.00		0.01

Field 2 - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,880.00	1,260.00	11,300.00		10.70
DL	100.00	100.00	100.00		0.01

Field 2 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	780.00	6,800.00		7.60
DL	100.00	100.00	100.00		0.01

Field 3 - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,710.00	1,080.00	11,800.00		7.78
DL	100.00	100.00	100.00		0.01

Field 3 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,910.00	900.00	4,030.00		3.40
DL	100.00	100.00	100.00		0.01

Geer - 10/14/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,350.00	840.00	6,840.00		5.33
DL	100.00	100.00	100.00		0.01

Geer - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,130.00	840.00	4,430.00		6.30
DL	100.00	100.00	100.00		0.01

George - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,540.00	1,120.00	11,200.00		9.01
DL	100.00	100.00	100.00		0.01

George - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,090.00	900.00	5,860.00		6.60
DL	100.00	100.00	100.00		0.01

Gratton - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,770.00	970.00	10,700.00		8.07
DL	100.00	100.00	100.00		0.01

Gratton - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	1,080.00	5,870.00		6.60
DL	100.00	100.00	100.00		0.01

Hake - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,080.00	12,800.00		7.17
DL	100.00	100.00	100.00		0.01

Hake - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	780.00	6,400.00		8.30
DL	100.00	100.00	100.00		0.01

Hickman - 10/14/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,590.00	890.00	7,100.00		5.05
DL	100.00	100.00	100.00		0.01

Hickman - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,960.00	960.00	3,680.00		4.40
DL	100.00	100.00	100.00		0.01

Quincy - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	750.00	8,940.00		5.81
DL	100.00	100.00	100.00		0.01

Quincy - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,820.00	920.00	4,880.00		4.10
DL	100.00	100.00	100.00		0.01

Swanson - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,240.00	820.00	5,160.00		6.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

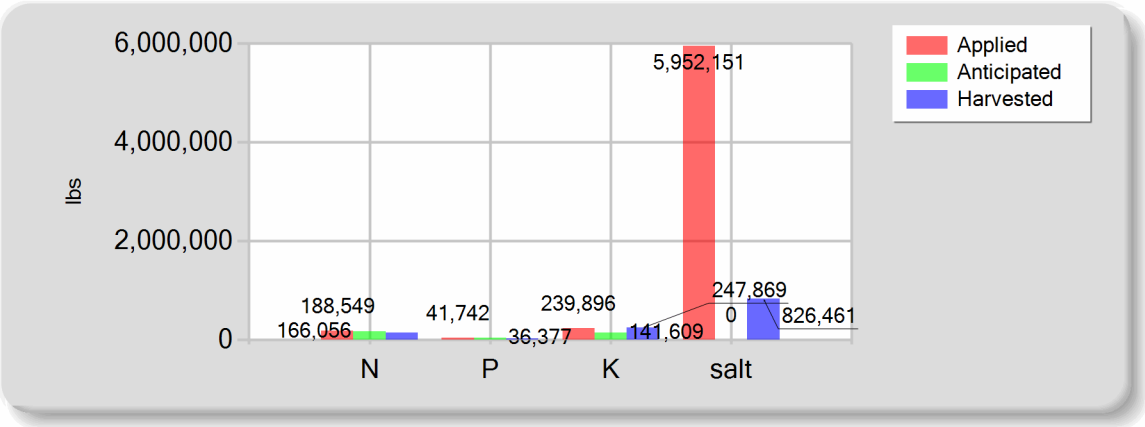
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

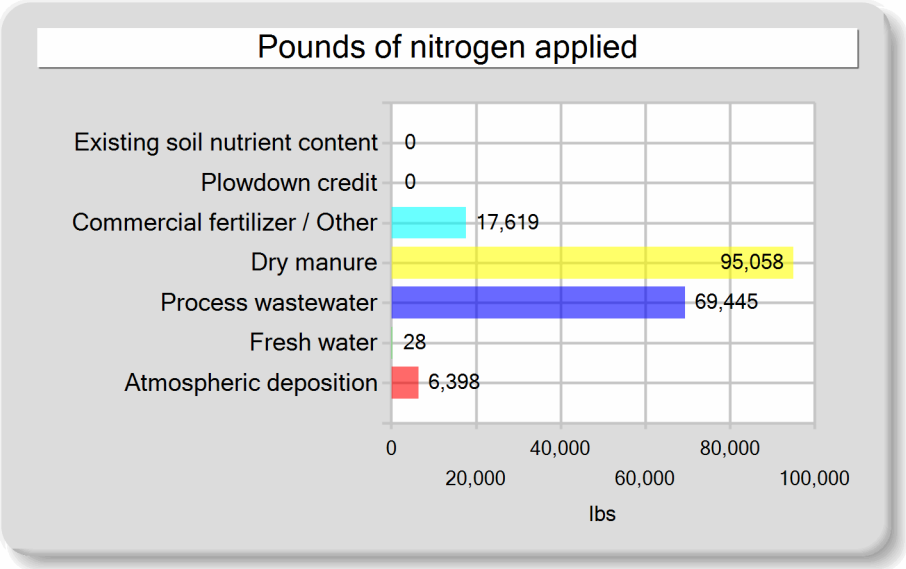
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

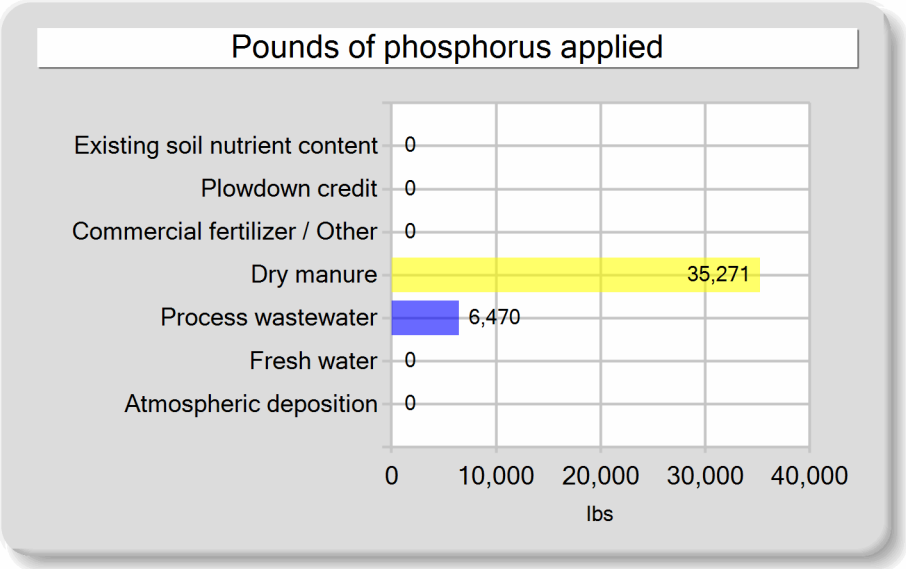
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	17,619.20	0.00	0.00	0.00
Dry manure	95,058.01	35,271.30	117,376.00	5,197,215.56
Process wastewater	69,445.10	6,470.25	122,519.59	690,392.00
Fresh water	28.24	0.00	0.00	64,542.98
Atmospheric deposition	6,398.00	0.00	0.00	0.00
Total nutrients applied	188,548.54	41,741.56	239,895.59	5,952,150.54
Anticipated crop nutrient removal	166,056.00	36,377.40	141,609.20	0.00
Actual crop nutrient removal	143,587.34	31,343.60	247,868.78	826,461.26
Nutrient balance	44,961.20	10,397.96	-7,973.19	5,125,689.28
Applied to removed ratio	1.31	1.33	0.97	7.20

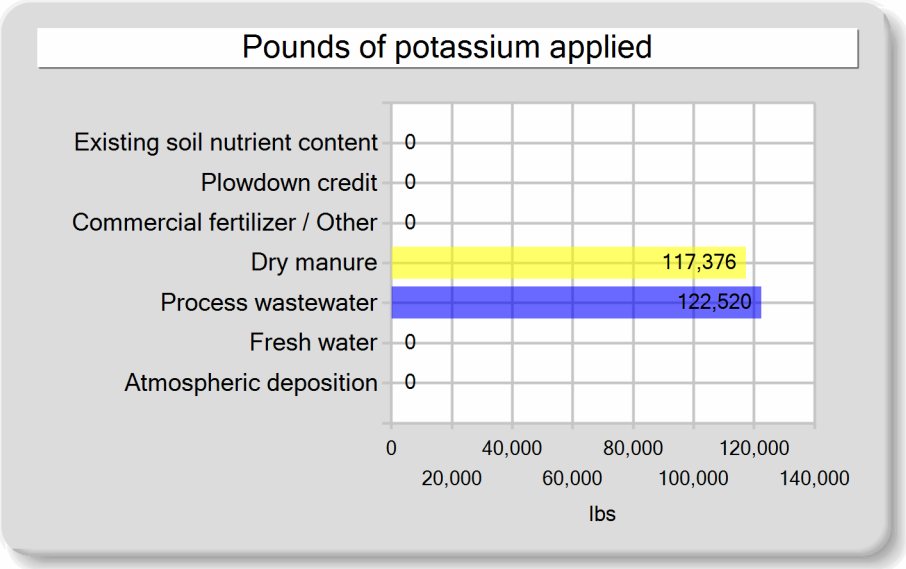
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

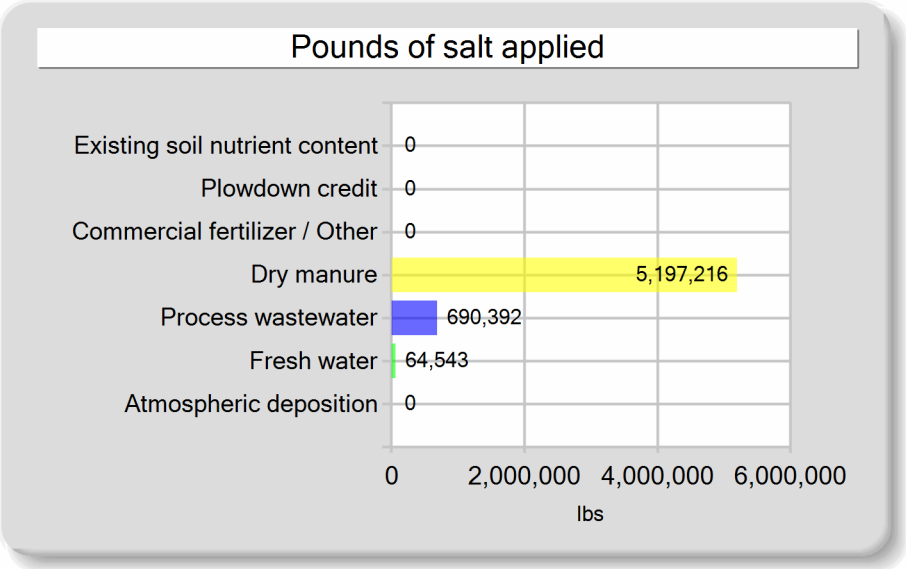


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Machado	Michael Machado
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-11-24	6-11-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
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www.denelelabs.com

Compliance Analysis Report

Customer
Machado & Sons Dairy-Denair
11706 E. Keyes Rd
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW Freestalls	T3355006A	Electrical Conductivity (EC)	EPA 120.1 and	0.655	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.5	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	355	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	18.8	mg/L
		Soluble Salts (SALT-SOL)	Calculation	419	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)			1.92
Electrical Conductivity (EC)	99.8			0.552 (T3354007B)	0.554 (Dup)	0.434
Nitrate Nitrogen (NO3-N)	103			18.5 (T3355006A)	19.8 (null)	7.06
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	102			0.280 (T3353003G)	0.280 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

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Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW Gratton	T3355006B	Electrical Conductivity (EC)	EPA 120.1 and	0.815	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	13.8	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	470	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	14.7	mg/L
		Soluble Salts (SALT-SOL)	Calculation	522	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)			1.92
Electrical Conductivity (EC)	99.8			0.552 (T3354007B)	0.554 (Dup)	0.434
Nitrate Nitrogen (NO3-N)	103			18.5 (T3355006A)	19.8 (null)	7.06
Total Dissolved Solids (TDS)	104			560 (T3362017B)	595 (Dup)	6.06
Total Nitrogen (TN)	102			0.280 (T3353003G)	0.280 (null)	0.000

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Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Gratton #2	T3355006C	Electrical Conductivity (EC)	EPA 120.1 and	0.623	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.46	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	398	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Dairy by Corrals	T3355006D	Electrical Conductivity (EC)	EPA 120.1 and	0.392	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.62	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	251	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Mikes House	T3355006E	Electrical Conductivity (EC)	EPA 120.1 and	0.384	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.37	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	246	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Dairy Yard	T3355006F	Electrical Conductivity (EC)	EPA 120.1 and	0.389	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.50	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	249	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Geer	T3355006G	Electrical Conductivity (EC)	EPA 120.1 and	0.922	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	25.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	590	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Machado & Sons Dairy-Denair
11706 E. Keyes Rd
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Hake	T3355006H	Electrical Conductivity (EC)	EPA 120.1 and	0.908	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	43.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	581	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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Compliance Analysis Report

Customer
Machado & Sons Dairy-Denair
11706 E. Keyes Rd
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Gratton	T3355006I	Electrical Conductivity (EC)	EPA 120.1 and	0.505	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.48	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	323	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.4	92.8 (T3355001A)	94.6 (T3355001A)	1.92
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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Compliance Analysis Report

Customer
Machado & Sons Dairy-Denair
11706 E. Keyes Rd
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Quincy	T3355006J	Electrical Conductivity (EC)	EPA 120.1 and	0.558	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	357	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.8	95.8 (T3355006J)	94.6 (T3355006J)	1.26
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: MACHADO + SONS
Mailing address:
(not facility address) DENAIR CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/20/23	3:00	IW FREESTALLS	AG WELL	16.4°c						H7
2		3:30	IW GRATTON	"	16.6°c						H7
3		3:40	DW-GRATTON #2	DOMESTIC	16.6°c		X				
4		3:50	DW-DAIRY BY CORRALS		16.0°c		X				
5		4:00	DW-MIKES HOUSE		16.6°c		X				
6		4:10	DW-DAIRY YARD		16.8°c		X				
7		4:20	DW-GEER		16.4°c		X				
8		4:30	DW-HAKE		15.8°c		X				
9		4:40	DW-GRATTON		15.6°c		X				
10		4:50	DW-QUINCY		16.0°c		X				

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	12-21-23 2:00			T3355006

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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Compliance Analysis Report

Customer
Machado & Sons Dairy-Denair
11706 E. Keyes Rd
Denair, CA 95316

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/21/23 2:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 12/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Hickman	T3355007A	Electrical Conductivity (EC)	EPA 120.1 and	0.449	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.84	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	288	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	91.8	95.8 (T3355006J)	94.6 (T3355006J)	1.26
Electrical Conductivity (EC)	99.8			0.434
Nitrate Nitrogen (NO3-N)	103			7.06

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: MACHADO + SONS
Mailing address:
(not facility address) DENAIR, CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES

Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
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SOIL
- K3 Soluble Phosphorus
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- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/20/23	5:00	DW- HICKMAN	DOMESTIC	16.6°C		X				
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	12-21-23 2:00			T3355007

Required to be submitted to any agency? (YES / NO)